

**TOFANIB 5MG FILM-COATED TABLETS**  
**Tofacitinib citrate**

**WARNING: SERIOUS INFECTIONS, MORTALITY, MALIGNANCY AND THROMBOSIS**

**SERIOUS INFECTIONS**

Patients treated with TOFANIB are at increased risk for developing serious infections that may lead to hospitalization or death. Most patients who developed these infections were taking concomitant immunosuppressants such as methotrexate or corticosteroids.

If a serious infection develops, interrupt TOFANIB until the infection is controlled.

Reported infections include:

- Active tuberculosis, which may present with pulmonary or extrapulmonary disease. Patients should be tested for latent tuberculosis before TOFANIB use and during therapy. Treatment for latent infection should be initiated prior to TOFANIB use.
- Invasive fungal infections, including cryptococcosis and pneumocystosis. Patients with invasive fungal infections may present with disseminated, rather than localized, disease.
- Bacterial, viral, including herpes zoster, and other infections due to opportunistic pathogens.

The risks and benefits of treatment with TOFANIB should be carefully considered prior to initiating therapy in patients with chronic or recurrent infection.

Patients should be closely monitored for the development of signs and symptoms of infection during and after treatment with TOFANIB, including the possible development of tuberculosis in patients who tested negative for latent tuberculosis infection prior to initiating therapy.

**MORTALITY**

Rheumatoid arthritis patients 50 years of age and older with at least one cardiovascular (CV) risk factor treated with TOFANIB 10 mg twice a day had a higher rate of all-cause mortality, including sudden CV death, compared to those treated with TOFANIB 5 mg given twice daily or TNF blockers in a large, ongoing, post marketing safety study.

**MALIGNANCIES**

Lymphoma and other malignancies have been observed in patients treated with TOFANIB. Epstein Barr Virus-associated post-transplant lymphoproliferative disorder has been observed at an increased rate in renal transplant patients treated with TOFANIB and concomitant immunosuppressive medications.

**THROMBOSIS**

Thrombosis, including pulmonary embolism, deep venous thrombosis, and arterial thrombosis, have occurred in patients treated with TOFANIB and other Janus kinase inhibitors used to treat inflammatory conditions. Rheumatoid arthritis patients who were 50 years of age and older with at least one CV risk factor treated with TOFANIB 10 mg twice daily compared to TOFANIB 5 mg twice daily or TNF blockers in a large, ongoing post marketing safety study had an observed increase in incidence of these events. Many of these events were serious and some resulted in death. Discontinue TOFANIB and promptly evaluate patients with symptoms of thrombosis.

## **1. NAME OF THE MEDICINAL PRODUCT**

TOFANIB 5MG FILM-COATED TABLETS

## **2. QUALITATIVE AND QUANTITATIVE COMPOSITION**

Each 5 mg film-coated tablet contains 8.08 mg of tofacitinib citrate equivalent to 5 mg of tofacitinib free base active pharmaceutical ingredient. Each 5 mg tablet also contains 121.32 mg lactose monohydrate.

## **3. PHARMACEUTICAL FORM**

*Film-coated tablets:*

White round film-coated tablets, show a white or off-white colour when the film coat is removed.

## **4. CLINICAL PARTICULARS**

### **4.1 Therapeutic indications**

#### **Rheumatoid Arthritis**

TOFANIB is indicated for the treatment of adult patients with moderately to severely active rheumatoid arthritis (RA) who have had an inadequate response or intolerance to methotrexate. It may be used as monotherapy or in combination with methotrexate or other non-biologic disease-modifying anti-rheumatic drugs (DMARDs).

Limitations of Use: Use of TOFANIB in combination with biologic DMARDs or potent immunosuppressants, such as azathioprine and cyclosporine is not recommended.

#### **Psoriatic Arthritis**

TOFANIB is indicated for the treatment of adult patients with active psoriatic arthritis (PsA) who have had an inadequate response or intolerance to methotrexate or other disease modifying antirheumatic drugs (DMARDs).

Limitations of Use: Use of TOFANIB in combination with biologic DMARDs or with potent immunosuppressants such as azathioprine and cyclosporine is not recommended.

#### **Ulcerative Colitis**

TOFANIB is indicated for the treatment of adult patients with moderately to severely active ulcerative colitis (UC), who have an inadequate response or who are intolerant to TNF blockers.

Limitations of Use: Use of TOFANIB in combination with biological therapies for UC or with potent immunosuppressants such as azathioprine and cyclosporine is not recommended.

### **4.2 Posology and method of administration**

#### **Important Administration Instructions**

- Do not initiate TOFANIB in patients with an absolute lymphocyte count less than 500 cells/mm<sup>3</sup>, an absolute neutrophil count (ANC) less than 1000 cells/mm<sup>3</sup> or who have hemoglobin levels less than 9 g/dL.
- Dose interruption is recommended for management of lymphopenia, neutropenia and anemia,
- Interrupt use of TOFANIB if a patient develops a serious infection until the infection is controlled.
- Take TOFANIB with or without food.

#### **Recommended Dosage in Rheumatoid Arthritis and Psoriatic Arthritis**

Table 1 displays the recommended adult daily dosage of TOFANIB and dosage adjustments for patients receiving CYP2C19 and/or CYP3A4 inhibitors, in patients with moderate or severe renal impairment

(including but not limited to those with severe insufficiency who are undergoing hemodialysis) or moderate hepatic impairment, with lymphopenia, neutropenia, or anemia.

**Table 1: Recommended Dosage of TOFANIB in Patients with Rheumatoid Arthritis and Psoriatic Arthritis<sup>1</sup>**

|                                                                                                                                                                                                                        | <b>TOFANIB</b>                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adult patients                                                                                                                                                                                                         | 5 mg twice daily                                                                                                                                                                                                                                            |
| Patients receiving: <ul style="list-style-type: none"> <li>strong CYP3A4 inhibitors (e.g., ketoconazole), or</li> <li>a moderate CYP3A4 inhibitor(s) with a strong CYP2C19 inhibitor(s) (e.g., fluconazole)</li> </ul> | 5 mg once daily                                                                                                                                                                                                                                             |
| Patients with: <ul style="list-style-type: none"> <li>moderate or severe renal impairment</li> <li>moderate hepatic impairment*</li> </ul>                                                                             | 5 mg once daily<br><br>For patients undergoing hemodialysis, dose should be administered after the dialysis session on dialysis days. If a dose was taken before the dialysis procedure, supplemental doses are not recommended in patients after dialysis. |
| Patients with lymphocyte count less than 500 cells/mm <sup>3</sup> , confirmed by repeat testing                                                                                                                       | Discontinue dosing.                                                                                                                                                                                                                                         |
| Patients with ANC 500 to 1000 cells/mm <sup>3</sup>                                                                                                                                                                    | Interrupt dosing.<br>When ANC is greater than 1000, resume 5 mg twice daily.                                                                                                                                                                                |
| Patients with ANC less than 500 cells/mm <sup>3</sup>                                                                                                                                                                  | Discontinue dosing.                                                                                                                                                                                                                                         |
| Patients with hemoglobin less than 8 g/dL or a decrease of more than 2 g/dL                                                                                                                                            | Interrupt dosing until hemoglobin values have normalized.                                                                                                                                                                                                   |

<sup>1</sup> TOFANIB is used in combination with nonbiologic disease modifying antirheumatic drugs (DMARDs) in psoriatic arthritis. The efficacy of TOFANIB as a monotherapy has not been studied in psoriatic arthritis.

\* Use of TOFANIB in patients with severe hepatic impairment is not recommended.

### **Recommended Dosage in Ulcerative Colitis**

Table 2 displays the recommended adult daily dosage of TOFANIB and dosage adjustments for patients receiving CYP2C19 and/or CYP3A4 inhibitors, with moderate or severe renal impairment (including but not limited to those with severe insufficiency who are undergoing hemodialysis) or moderate hepatic impairment, with lymphopenia, neutropenia or anemia.

**Table 2: Recommended Dosage of TOFANIB in Patients with UC**

|                | <b>TOFANIB</b>                                                                                                                                                                                                       |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adult patients | <b>Induction:</b> 10 mg twice daily for at least 8 weeks; evaluate patients and transition to maintenance therapy depending on therapeutic response. If needed continue 10 mg twice daily for a maximum of 16 weeks. |

|                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                   | <p>Discontinue 10 mg twice daily after 16 weeks if adequate therapeutic response is not achieved.</p> <p><b>Maintenance:</b> 5 mg twice daily.</p> <p>For patients with loss of response during maintenance treatment, a dosage of 10 mg twice daily may be considered and limited to the shortest duration, with careful consideration of the benefits and risks for the individual patient. Use the lowest effective dose needed to maintain response.</p> |
| <p>Patients receiving:</p> <ul style="list-style-type: none"> <li>• Strong CYP3A4 inhibitors (e.g., ketoconazole), or</li> <li>• a moderate CYP3A4 inhibitor(s) with a strong CYP2C19 inhibitor(s) (e.g., fluconazole)</li> </ul> | <p>If taking 10 mg twice daily, reduce to 5 mg twice daily.</p> <p>If taking 5 mg twice daily, reduce to 5 mg once daily.</p>                                                                                                                                                                                                                                                                                                                                |
| <p>Patients with:</p> <ul style="list-style-type: none"> <li>• moderate or severe renal impairment</li> <li>• moderate hepatic impairment</li> </ul>                                                                              | <p>If taking 10 mg twice daily, reduce to 5 mg twice daily.</p> <p>If taking 5 mg twice daily, reduce to 5 mg once daily.</p>                                                                                                                                                                                                                                                                                                                                |
|                                                                                                                                                                                                                                   | <p>For patients undergoing hemodialysis, dose should be administered after the dialysis session on dialysis days. If a dose was taken before the dialysis procedure, supplemental doses are not recommended in patients after dialysis.</p>                                                                                                                                                                                                                  |
| Patients with lymphocyte count less than 500 cells/mm <sup>3</sup> , confirmed by repeat testing                                                                                                                                  | Discontinue dosing.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Patients with ANC 500 to 1000 cells/mm <sup>3</sup>                                                                                                                                                                               | <p>If taking 10 mg twice daily, reduce to 5 mg twice daily. When ANC is greater than 1000, increase to 10 mg twice daily based on clinical response.</p> <p>If taking 5 mg twice daily, interrupt dosing. When ANC is greater than 1000, resume 5 mg twice daily.</p>                                                                                                                                                                                        |
| Patients with ANC less than 500 cells/mm <sup>3</sup>                                                                                                                                                                             | Discontinue dosing                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Patients with hemoglobin less than 8 g/dL or a decrease of more than 2 g/dL                                                                                                                                                       | Interrupt dosing until hemoglobin values have normalized.                                                                                                                                                                                                                                                                                                                                                                                                    |

\*Use of TOFANIB in patients with severe hepatic impairment is not recommended.

## **Hepatic Impairment**

### Severe Impairment

TOFANIB has not been studied in patients with severe hepatic impairment; therefore, use of TOFANIB in patients with severe hepatic impairment is not recommended.

### Moderate Impairment

TOFANIB-treated patients with moderate hepatic impairment had greater tofacitinib blood concentration than TOFANIB-treated patients with normal hepatic function. Higher blood concentrations may increase the risk of some adverse reactions. Therefore, dosage adjustment of TOFANIB is recommended in patients with moderate hepatic impairment.

### Mild Impairment

No dosage adjustment of TOFANIB is required in patients with mild hepatic impairment.

### Hepatitis B or C Serology

The safety and efficacy of TOFANIB have not been studied in patients with positive hepatitis B virus or hepatitis C virus serology.

## **Renal Impairment**

### Moderate and Severe Impairment

TOFANIB-treated patients with moderate or severe renal impairment had greater tofacitinib blood concentrations than TOFANIB-treated patients with normal renal function. Therefore, dosage adjustment of TOFANIB is recommended in patients with moderate or severe renal impairment (including but not limited to those with severe insufficiency who are undergoing hemodialysis).

### Mild Impairment

No dosage adjustment is required in patients with mild renal impairment.

## **Geriatric Use**

Of the 3315 patients who enrolled in rheumatoid arthritis Studies I to V, a total of 505 rheumatoid arthritis patients were 65 years of age and older, including 71 patients 75 years and older. The frequency of serious infection among tofacitinib-treated subjects 65 years of age and older was higher than among those under the age of 65.

Of the 1156 TOFANIB treated patients in the UC program, a total of 77 patients (7%) were 65 years of age or older. The number of patients aged 65 years and older was not sufficient to determine whether they responded differently from younger patients.

As there is a higher incidence of infections in the elderly population in general, caution should be used when treating the elderly.

## **Pediatric Use**

The safety and effectiveness of TOFANIB in pediatric patients have not been established.

## **Use in Diabetics**

As there is a higher incidence of infection in diabetic population in general, caution should be used when treating patients with diabetes.

## **4.3 Contraindications**

Hypersensitivity to the active substance or to any of the excipients.

#### **4.4 Special warnings and precautions for use**

##### **Serious Infections**

Serious and sometimes fatal infections due to bacterial, mycobacterial, invasive fungal, viral, or other opportunistic pathogens have been reported in patients receiving immunomodulatory agents, including biologic DMARDs and TOFANIB. The most common serious infections reported with TOFANIB included pneumonia, cellulitis, herpes zoster, urinary tract infection, diverticulitis, and appendicitis. Among opportunistic infections, tuberculosis and other mycobacterial infections, cryptococcus, histoplasmosis, esophageal candidiasis, multidermatomal herpes zoster, cytomegalovirus infection, BK virus infections, and listeriosis were reported with TOFANIB. Some patients have presented with disseminated rather than localized disease, and rheumatoid arthritis patients were often taking concomitant immunomodulating agents such as methotrexate or corticosteroids which, in addition to rheumatoid arthritis may predispose them to infections. Other serious infections that were not reported in clinical studies, may also occur (e.g., coccidioidomycosis).

In the UC population, TOFANIB treatment with 10 mg twice daily was associated with greater risk of serious infections compared to 5 mg twice daily. Additionally, opportunistic herpes zoster infections (including meningoencephalitis, ophthalmologic, and disseminated cutaneous) were seen in patients who were treated with TOFANIB 10 mg twice daily.

TOFANIB should not be initiated in patients with an active infection, including localized infections. The risks and benefits of treatment should be considered prior to initiating TOFANIB in patients with chronic or recurrent infections, or those who have been exposed to tuberculosis, or with a history of a serious or an opportunistic infection, or have resided or travelled in areas of endemic tuberculosis or endemic mycoses; or have underlying conditions that may predispose them to infection.

Patients should be closely monitored for the development of signs and symptoms of infection during and after treatment with TOFANIB. TOFANIB should be interrupted if a patient develops a serious infection, an opportunistic infection, or sepsis. A patient who develops a new infection during treatment with TOFANIB should undergo prompt and complete diagnostic testing appropriate for an immunocompromised patient, appropriate antimicrobial therapy should be initiated, and the patient should be closely monitored.

As there is a higher incidence of infections in the elderly and in the diabetic populations in general, caution should be used when treating the elderly and patients with diabetes (see section 4.2). Caution is also recommended in patients with a history of chronic lung disease as they may be more prone to infections. Events of interstitial lung disease (some of which had a fatal outcome) have been reported in patients treated with TOFANIB, a Janus-kinase (JAK) inhibitor, in clinical trials and in the post-marketing setting although the role of JAK inhibition in these events is not known.

Risk of infection may be higher with increasing degrees of lymphopenia and consideration should be given to lymphocyte counts when assessing individual patient risk of infection. Discontinuation and monitoring criteria for lymphopenia are discussed in *section 4.2*.

##### **Tuberculosis**

Patients should be evaluated and tested for latent or active infection prior to and per applicable guidelines during administration of TOFANIB.

Patients with latent tuberculosis should be treated with standard antimycobacterial therapy before administering TOFANIB.

Antituberculosis therapy should also be considered prior to administration of TOFANIB in patients with a past history of latent or active tuberculosis in whom an adequate course of treatment cannot be confirmed,

and for patients with a negative test for latent tuberculosis but who have risk factors for tuberculosis infection. Consultation with a health care professional with expertise in the treatment of tuberculosis is recommended to aid in the decision about whether initiating antituberculosis therapy is appropriate for an individual patient.

Patients should be closely monitored for the development of signs and symptoms of tuberculosis, including patients who tested negative for latent tuberculosis infection prior to initiating therapy.

### **Viral Reactivation**

Viral reactivation has been reported with DMARD treatment and cases of herpes virus reactivation (e.g., herpes zoster) were observed. Post-marketing cases of hepatitis B reactivation have been reported in patients treated with TOFANIB. The impact of TOFANIB on chronic viral hepatitis reactivation is unknown. Patients who screened positive for hepatitis B or C were excluded from clinical trials. Screening for viral hepatitis should be performed in accordance with clinical guidelines before starting therapy with TOFANIB. The risk of herpes zoster appears to be higher in Japanese and Korean patients treated with TOFANIB.

### **Venous Thromboembolism**

Venous thromboembolism (VTE) has been observed in patients taking TOFANIB in clinical trials and post-marketing reporting. In one large ongoing randomized post authorization safety surveillance (PASS) study in RA patients who were 50 years or older with at least one cardiovascular risk factor, patients were treated with tofacitinib 5 mg twice daily, tofacitinib 10 mg twice daily or a TNF inhibitor. A dose dependent increase in pulmonary embolism (PE) events was observed in patients treated with tofacitinib compared to TNF inhibitors. Many of these PE events were serious and some resulted in death. PE events were reported more frequently in this study in patients taking tofacitinib relative to other studies across the tofacitinib program.

Deep vein thrombosis (DVT) events were observed in all three treatment groups in this study.

Assess patients for VTE risk factors before starting treatment and periodically during treatment. Use TOFANIB with caution in patients in whom risk factors are identified. Urgently evaluate patients with signs and symptoms of VTE. Discontinue tofacitinib while evaluating suspected VTE, regardless of dose or indication.

### **Mortality**

Rheumatoid arthritis patients 50 years of age and older with at least one cardiovascular (CV) risk factor treated with TOFANIB 10 mg twice a day had a higher rate of all-cause mortality, including sudden CV death, compared to those treated with TOFANIB 5 mg given twice daily or TNF blockers in a large, ongoing, post marketing safety study.

A dosage of TOFANIB 10 mg twice daily is not recommended for the treatment of RA or PsA.

For the treatment of UC, use TOFANIB at the lowest effective dose and for the shortest duration needed to achieve/maintain therapeutic response.

### **Malignancy and Lymphoproliferative Disorder (Excluding Non-melanoma Skin Cancer [NMSC])**

Consider the risks and benefits of TOFANIB treatment prior to initiating therapy in patients with current or a history of malignancy other than a successfully treated non-melanoma skin cancer (NMSC) or when considering continuing TOFANIB in patients who develop a malignancy. The possibility exists for TOFANIB to affect host defenses against malignancies.

Lymphomas have been observed in patients treated with TOFANIB. Patients with rheumatoid arthritis, particularly those with highly active disease may be at a higher risk (up to several-fold) than the general population for the development of lymphoma. The role of TOFANIB, in the development of lymphoma is uncertain.

Other malignancies were observed in clinical studies and the post-marketing setting, including, but not limited to, lung cancer, breast cancer, melanoma, prostate cancer and pancreatic cancer.

The role of treatment with TOFANIB on the development and course of malignancies is not known.

Recommendations for non-melanoma skin cancer are presented below.

#### Rheumatoid Arthritis

In controlled Phase 3 clinical studies in rheumatoid arthritis patients, 26 malignancies (excluding NMSC) including 5 lymphoma were diagnosed in 26 patients receiving TOFANIB/TOFANIB plus DMARD, compared to 0 malignancies (excluding NMSC) in patients in the placebo/placebo plus DMARD and 2 in 2 patients in the adalimumab group, 1 in 1 patient in the methotrexate group. 3800 patients (3942 patient-years of observation) were treated with TOFANIB for durations up to 2 years while 681 patients (203 patient-years of observation) were treated with placebo for a maximum of 6 months and 204 patients (179 patient-years of observation) were treated with adalimumab for 12 months. The exposure-adjusted incidence rate for malignancies and lymphoma was 0.66 and 0.13 events per 100 patient-years, respectively, in the TOFANIB groups.

In the long-term safety population (4867 patients), in rheumatoid arthritis studies, the rate of malignancies (excluding NMSC) and lymphoma was 0.97 and 0.09 events per 100 patient-years respectively, consistent with the rate observed in the controlled period.

#### Psoriatic Arthritis

During the 2 PsA controlled clinical studies there were 3 malignancies (excluding NMSC) in 474 patients receiving TOFANIB plus nonbiologic DMARD (6 to 12 months exposure) compared with 0 malignancies in 236 patients in the placebo plus nonbiologic DMARD group (3 months exposure) and 0 malignancies in 106 patients in the adalimumab plus nonbiologic DMARD group (12 months exposure). No lymphomas were reported. Malignancies have also been observed in the long term extension study in psoriatic arthritis patients treated with TOFANIB.

#### Ulcerative Colitis

All information provided in this section for the ulcerative colitis indication is applicable to TOFANIB 5 mg and 10 mg twice daily as they contain the same active ingredient (tofacitinib).

In the placebo-controlled induction and maintenance studies for ulcerative colitis, there were no malignancies (excluding NMSC) in any TOFANIB group. In the entire TOFANIB treatment experience for ulcerative colitis, malignancies (excluding NMSC) have been reported with an overall incidence rate of 0.5 events per 100 patient-years.

#### **Non-melanoma Skin Cancer**

Non-melanoma skin cancers (NMSCs) have been reported in patients treated with TOFANIB. Periodic skin examination is recommended for patients who are at increased risk for skin cancer. In the UC population, treatment with TOFANIB 10 mg twice daily was associated with greater risk of NMSC.



### **Thrombosis**

Thrombosis, including pulmonary embolism, deep venous thrombosis, and arterial thrombosis, have occurred in patients treated with TOFANIB and other Janus kinase (JAK) inhibitors used to treat inflammatory conditions. Patients with rheumatoid arthritis 50 years of age and older with at least one CV risk factor treated with TOFANIB 10 mg twice daily compared to TOFANIB 5 mg twice daily or TNF blockers in a large, ongoing postmarketing study had an observed increase in incidence of these events. Many of these events were serious and some resulted in death.

A dosage of TOFANIB 10 mg twice daily is not recommended for the treatment of RA.

In a long-term extension study in patients with UC, four cases of pulmonary embolism were reported in patients taking TOFANIB 10 mg twice daily, including one death in a patient with advanced cancer.

Promptly evaluate patients with symptoms of thrombosis and discontinue TOFANIB in patients with symptoms of thrombosis.

Avoid TOFANIB in patients that may be at increased risk of thrombosis. For the treatment of UC, use TOFANIB at the lowest effective dose and for the shortest duration needed to achieve/maintain therapeutic response.

### **Gastrointestinal Perforations**

Events of gastrointestinal perforation have been reported in clinical trials. The role of JAK inhibition in these events is not known. Events were primarily reported as diverticular perforation, peritonitis, abdominal abscess and appendicitis. In the rheumatoid arthritis clinical trials, the incidence rate of gastrointestinal perforation across all studies (Phase 1, Phase 2, Phase 3 and long term extension) for all treatments groups all doses was 0.11 events per 100 patient-years with TOFANIB therapy. Rheumatoid arthritis patients who developed gastrointestinal perforations were taking concomitant non-steroidal anti-inflammatory drugs (NSAIDs) and/or corticosteroids. The relative contribution of these concomitant medications vs. TOFANIB to the development of gastrointestinal perforations is not known.

TOFANIB should be used with caution in patients who may be at increased risk for gastrointestinal perforation (e.g., patients with a history of diverticulitis). Patients presenting with new onset abdominal symptoms should be evaluated promptly for early identification of gastrointestinal perforation. There was no discernable difference in frequency of gastrointestinal perforation between the placebo and the TOFANIB arms in clinical trials of patients with UC, and many of them were receiving background corticosteroids.

### **Hypersensitivity**

Reactions such as angioedema and urticaria that may reflect drug hypersensitivity have been observed in patients receiving TOFANIB. Some events were serious. Many of these events occurred in patients that have a history of multiple allergies. If a serious hypersensitivity reaction occurs, promptly discontinue TOFANIB while evaluating the potential cause or causes of the reaction.

### **Laboratory Parameters**

*Lymphocytes:* Lymphocyte counts <500 cells/mm<sup>3</sup> were associated with an increased incidence of treated and serious infections. It is not recommended to initiate TOFANIB treatment in patients with a low lymphocyte count (i.e., <500 cells/mm<sup>3</sup>). In patients who develop a confirmed absolute lymphocyte count <500 cells/mm<sup>3</sup> treatment with TOFANIB is not recommended. Lymphocytes should be monitored at baseline and every 3 months thereafter. For recommended modifications based on lymphocyte counts.

*Neutrophils:* Treatment with TOFANIB was associated with an increased incidence of neutropenia ( $<2000$  cells/mm<sup>3</sup>) compared to placebo. It is not recommended to initiate TOFANIB treatment in patients with a low neutrophil count (i.e., ANC  $<1000$  cells/mm<sup>3</sup>). For patients who develop a persistent ANC of 500-1000 cells/mm<sup>3</sup>, reduce TOFANIB dose or interrupt TOFANIB dosing until ANC is  $>1000$  cells/mm<sup>3</sup>. In patients who develop a confirmed absolute neutrophil count  $<500$  cells/mm<sup>3</sup>, treatment with TOFANIB is not recommended. Neutrophils should be monitored at baseline and after 4 to 8 weeks of treatment and every 3 months thereafter.

*Hemoglobin:* It is not recommended to initiate TOFANIB treatment in patients with low hemoglobin values (i.e.,  $<9$  g/dL). Treatment with TOFANIB should be interrupted in patients who develop hemoglobin levels  $<8$  g/dL or whose hemoglobin level drops  $>2$  g/dL on treatment. Hemoglobin should be monitored at baseline and after 4 - 8 weeks of treatment and every 3 months thereafter.

*Lipids:* Treatment with TOFANIB was associated with increases in lipid parameters such as total cholesterol, low-density lipoprotein (LDL) cholesterol, and high-density lipoprotein (HDL) cholesterol. Maximum effects were generally observed within 6 weeks. Assessment of lipid parameters should be performed approximately 4 - 8 weeks following initiation of TOFANIB therapy. Patients should be managed according to current local clinical guidelines for the management of hyperlipidemia. Increase in total and LDL cholesterol associated with TOFANIB may be decreased to pre-treatment levels with statin therapy.

In the clinical trials in psoriatic arthritis, ankylosing spondylitis, ulcerative colitis, and polyarticular course juvenile idiopathic arthritis changes in lymphocytes, neutrophils, and lipids observed with TOFANIB treatment were similar to the changes observed in clinical trials in rheumatoid arthritis.

### **Vaccinations**

No data are available on the secondary transmission of infection by live vaccines to patients receiving TOFANIB. It is recommended that live vaccines not be given concurrently with TOFANIB. It is recommended that all patients be brought up to date with all immunizations in agreement with current immunization guidelines prior to initiating TOFANIB therapy. The interval between live vaccinations and initiation of TOFANIB therapy should be in accordance with current vaccination guidelines regarding immunomodulatory agents. Consistent with these guidelines, if live zoster vaccine is administered, it should only be administered to patients with a known history of chickenpox or those that are seropositive for varicella zoster virus. Vaccination should occur at least 2 weeks but preferably 4 weeks before initiating immunomodulatory agents such as TOFANIB.

A controlled study in patients with rheumatoid arthritis on background methotrexate evaluated the humoral and cell-mediated responses to immunization with a live attenuated virus vaccine (Zostavax) indicated for prevention of herpes zoster. The immunization occurred 2 to 3 weeks before initiating a 12-week treatment with TOFANIB 5 mg twice daily or placebo. Six weeks after immunization with the zoster vaccine, TOFANIB and placebo recipients exhibited similar humoral and cell-mediated responses (mean fold change of VZV IgG antibodies 2.11 in TOFANIB 5 mg twice daily and 1.74 in placebo twice daily; VZV IgG fold-rise  $\geq 1.5$  in 57% of TOFANIB recipients and in 43% of placebo recipients; mean fold change of VZV T-cell ELISPOT Spot Forming Cells 1.5 in TOFANIB 5 mg twice daily and 1.29 in placebo twice daily). These responses were similar to those observed in healthy volunteers aged 50 years and older.

In this study one patient experienced dissemination of the vaccine strain of varicella zoster virus, 16 days after vaccination. The patient was varicella virus naïve, as evidenced by no previous history of varicella infection and no anti-varicella antibodies at baseline. TOFANIB was discontinued and the subject recovered after treatment with standard doses of antiviral medication. Subsequent testing showed that this patient made robust anti-varicella T-cell and antibody responses to the vaccine approximately 6 weeks post-vaccination, but not at 2 weeks post-vaccination, as expected for a primary infection.

## 4.5 Fertility, pregnancy and lactation

### Pregnancy

#### Risk Summary

Available data with TOFANIB use in pregnant women are insufficient to establish a drug associated risk of major birth defects, miscarriage or adverse maternal or fetal outcomes. There are risks to the mother and the fetus associated with rheumatoid arthritis and UC in pregnancy. In animal reproduction studies, fetocidal and teratogenic effects were noted when pregnant rats and rabbits received tofacitinib during the period of organogenesis at exposures multiples of 73 times. Further, in a peri and post-natal study in rats, tofacitinib resulted in reductions in live litter size, postnatal survival, and pup body weights at exposure multiples of approximately 73 times the recommended dose of 5 mg twice daily.

Women of reproductive potential should be advised to use effective contraception during treatment with tofacitinib and for at least 4 weeks after the last dose.

The estimated background risks of major birth defects and miscarriage for the indicated populations are unknown. All pregnancies have a background risk of birth defect, loss, or other adverse outcomes. The background risks in the U.S. general population of major birth defects and miscarriages are 2 to 4% and 15 to 20% of clinically recognized pregnancies, respectively.

#### Clinical Considerations

##### *Disease-associated Maternal and/or Embryo/Fetal Risk*

Published data suggest that increased disease activity is associated with the risk of developing adverse pregnancy outcomes in women with rheumatoid arthritis or ulcerative colitis. Adverse pregnancy outcomes include preterm delivery (before 37 weeks of gestation), low birth weight (less than 2500 g) infants, and small for gestational age at birth.

#### Data

##### *Animal Data*

In a rat embryofetal developmental study, in which pregnant rats received tofacitinib during organogenesis, tofacitinib was teratogenic at exposure levels approximately 146 times the recommended dose of 5 mg twice daily (on an AUC basis at oral doses of 100 mg/kg/day in rats). Teratogenic effects consisted of external and soft tissue malformations of anasarca and membranous ventricular septal defects, respectively, and skeletal malformations or variations (absent cervical arch; bent femur, fibula, humerus, radius, scapula, tibia, and ulna; sternoschisis; absent rib; misshapen femur; branched rib; fused rib; fused sternebra; and hemicentric thoracic centrum). In addition, there was an increase in post-implantation loss, consisting of early and late resorptions, resulting in a reduced number of viable fetuses. Mean fetal body weight was reduced. No developmental toxicity was observed in rats at exposure levels approximately 58 times the recommended dose of 5 mg twice daily (on an AUC basis at oral doses of 30 mg/kg/day in pregnant rats).

In a rabbit embryofetal developmental study, in which pregnant rabbits received tofacitinib during the period of organogenesis, tofacitinib was teratogenic at exposure levels approximately 13 times the recommended dose of 5 mg twice daily (on an AUC basis at oral doses of 30 mg/kg/day in rabbits) in the absence of signs of maternal toxicity. Teratogenic effects included thoracogastroschisis, omphalocele, membranous ventricular septal defects, and cranial/skeletal malformations (microstomia, microphthalmia), mid-line and tail defects. In addition, there was an increase in post-implantation loss associated with late resorptions. No developmental toxicity was observed in rabbits at exposure levels approximately 3 times the recommended dose of 5 mg twice daily (on an AUC basis at oral doses of 10 mg/kg/day in pregnant rabbits).

In a peri- and postnatal development study in pregnant rats that received tofacitinib from gestation day 6 through day 20 of lactation, there were reductions in live litter size, postnatal survival, and pup body weights

at exposure levels approximately 73 times the recommended dose of 5 mg twice daily (on an AUC basis at oral doses of 50 mg/kg/day in rats). There was no effect on behavioral and learning assessments, sexual maturation or the ability of the F1 generation rats to mate and produce viable F2 generation fetuses in rats at exposure levels approximately 17 times the recommended dose of 5 mg twice daily (on an AUC basis at oral doses of 10 mg/kg/day in rats).

## **Lactation**

### Risk Summary

There are no data on the presence of tofacitinib in human milk, the effects on a breastfed infant, or the effects on milk production. Tofacitinib is present in the milk of lactating rats. When a drug is present in animal milk, it is likely that the drug will be present in human milk. Given the serious adverse reactions seen in patients treated with TOFANIB, such as increased risk of serious infections, advise patients that breastfeeding is not recommended during treatment and for at least 18 hours after the last dose of TOFANIB.

### Data

Following administration of tofacitinib to lactating rats, concentrations of tofacitinib in milk over time paralleled those in serum, and were approximately 2 times higher in milk relative to maternal serum at all time points measured.

## **Females and Males of Reproductive Potential**

### Contraception

#### *Females*

In an animal reproduction study, tofacitinib at AUC multiples of 13 times the recommended dose of 5 mg twice daily demonstrated adverse embryo-fetal findings.

However, there is uncertainty as to how these animal findings relate to females of reproductive potential treated with the recommended clinical dose. Consider pregnancy planning and prevention for females of reproductive potential.

### Infertility

#### *Females*

Based on findings in rats, treatment with TOFANIB may result in reduced fertility in females of reproductive potential. It is not known if this effect is reversible.

## **4.6 Effects on ability to drive and use machines**

No formal studies have been conducted on the effects on the ability to drive and use machines.

## **4.7 Adverse reactions**

The following clinically significant adverse reactions are described elsewhere in the labeling:

- Serious Infections
- Mortality
- Malignancy and Lymphoproliferative Disorders
- Thrombosis
- Gastrointestinal Perforations
- Hypersensitivity
- Laboratory Abnormalities

## **Clinical Trials Experience**

Because clinical studies are conducted under widely varying conditions, adverse reaction rates observed in the clinical studies of a drug cannot be directly compared to rates in the clinical studies of another drug and may not predict the rates observed in a broader patient population in clinical practice.

## **Rheumatoid Arthritis**

The clinical studies described in the following sections were conducted using TOFANIB. Although other doses of TOFANIB have been studied, the recommended dose of TOFANIB is 5 mg twice daily.

The following data includes two Phase 2 and five Phase 3 double-blind, controlled, multicenter trials. In these trials, patients were randomized to doses of TOFANIB 5 mg twice daily (292 patients) and 10 mg twice daily (306 patients) monotherapy, TOFANIB 5 mg twice daily (1044 patients) and 10 mg twice daily (1043 patients) in combination with DMARDs (including methotrexate) and placebo (809 patients). All seven protocols included provisions for patients taking placebo to receive treatment with TOFANIB at Month 3 or Month 6 either by patient response (based on uncontrolled disease activity) or by design, so that adverse events cannot always be unambiguously attributed to a given treatment. Therefore, some analyses that follow include patients who changed treatment by design or by patient response from placebo to TOFANIB in both the placebo and TOFANIB group of a given interval. Comparisons between placebo and TOFANIB were based on the first 3 months of exposure, and comparisons between TOFANIB 5 mg twice daily and TOFANIB 10 mg twice daily were based on the first 12 months of exposure.

The long-term safety population includes all patients who participated in a double-blind, controlled trial (including earlier development phase studies) and then participated in one of two long-term safety studies. The design of the long-term safety studies allowed for modification of TOFANIB doses according to clinical judgment. This limits the interpretation of the long-term safety data with respect to dose.

The most common serious adverse reactions were serious infections.

The proportion of patients who discontinued treatment due to any adverse reaction during the 0 to 3 months exposure in the double-blind, placebo-controlled trials was 4% for patients taking TOFANIB and 3% for placebo-treated patients.

Safety information from ad hoc interim analyses is also included for one large (N=4362), ongoing randomized post authorization safety surveillance (PASS) study in RA patients who were 50 years or older with at least one cardiovascular risk factor (CV risk factors defined as: current cigarette smoker, diagnosis of hypertension, diabetes mellitus, family history of premature coronary heart disease, history of coronary artery disease including a history of revascularization procedure, coronary artery bypass grafting, myocardial infarction, cardiac arrest, unstable angina, acute coronary syndrome, and presence 15 of extra articular disease associated with RA, e.g. nodules, Sjögren's syndrome, anemia of chronic disease, pulmonary manifestations), and were on a stable background dose of methotrexate.

Patients were randomized to open-label tofacitinib 10 mg twice daily, tofacitinib 5 mg twice daily, or a TNF inhibitor (TNF inhibitor was either etanercept 50 mg once weekly or adalimumab 40 mg every other week) in a 1:1:1 ratio. The co-primary endpoints are adjudicated malignancy (excluding NMSC) and adjudicated major adverse cardiovascular events (MACE); cumulative incidence and statistical assessment of endpoints are blinded. The study is an event-powered study that also requires at least 1500 patients to be followed for 3 years. The study treatment of tofacitinib 10 mg twice daily has been stopped and the patients were switched to 5 mg twice daily because of a dose dependent signal of PE.

## **Overall Infections**

In the seven controlled trials, during the 0 to 3 months exposure, the overall frequency of infections was 20% and 22% in the 5 mg twice daily and 10 mg twice daily groups, respectively, and 18% in the placebo group.

The most commonly reported infections with TOFANIB were upper respiratory tract infections, nasopharyngitis, and urinary tract infections (4%, 3%, and 2% of patients, respectively).

### Ulcerative Colitis

In the randomized 8 week Phase 2/3 induction studies, the proportions of patients with infections were 21.1% for TOFANIB 10 mg twice daily compared with 15.2% for placebo. In the randomized 52-week Phase 3 maintenance study, the proportion of patients with infections were 35.9% for TOFANIB 5 mg twice daily, 39.8% for TOFANIB 10 mg twice daily, and 24.2% for placebo. In the entire treatment experience with TOFANIB in the ulcerative colitis program, the overall incidence rate of infection was 65.7 events per 100 patient years (involving 47.9% of patients). The most common infection was nasopharyngitis, occurring in 16.8% of patients.

### **Serious Infections**

In the seven controlled trials, during the 0 to 3 months exposure, serious infections were reported in 1 patient (0.5 events per 100 patient-years) who received placebo and 11 patients (1.7 events per 100 patient-years) who received TOFANIB 5 mg or 10 mg twice daily. The rate difference between treatment groups (and the corresponding 95% confidence interval) was 1.1 (-0.4, 2.5) events per 100 patient-years for the combined 5 mg twice daily and 10 mg twice daily TOFANIB group minus placebo. In the seven controlled trials, during the 0 to 12 months exposure, serious infections were reported in 34 patients (2.7 events per 100 patient-years) who received 5 mg twice daily of TOFANIB and 33 patients (2.7 events per 100 patient-years) who received 10 mg twice daily of TOFANIB. The rate difference between TOFANIB doses (and the corresponding 95% confidence interval) was -0.1 (-1.3, 1.2) events per 100 patient-years for 10 mg twice daily TOFANIB minus 5 mg twice daily TOFANIB. The most common serious infections included pneumonia, cellulitis, herpes zoster, and urinary tract infection.

### Ulcerative Colitis

TOFANIB has been studied in patients with moderately to severely active UC in 4 randomized, double-blind, placebo-controlled trials (UC-I, UC-II, UC-III, and dose-ranging UC-V) and an open-label long-term extension study (UC-IV).

Adverse reactions reported in  $\geq 5\%$  of patients treated with either 5 mg or 10 mg twice daily of TOFANIB and  $\geq 1\%$  greater than reported in patients receiving placebo in either the induction or maintenance clinical trials were: nasopharyngitis, elevated cholesterol levels, headache, upper respiratory tract infection, increased blood creatine phosphokinase, rash, diarrhea, and herpes zoster.

#### *Induction Trials (Study UC-I, UC-II, and UC-V):*

Common adverse reactions reported in  $\geq 2\%$  of patients treated with TOFANIB 10 mg twice daily and  $\geq 1\%$  greater than that reported in patients receiving placebo in the 3 induction trials were: headache, nasopharyngitis, elevated cholesterol levels, acne, increased blood creatine phosphokinase, and pyrexia.

#### *Maintenance Trial (Study UC-III)*

Common adverse reactions reported in  $\geq 4\%$  of patients treated with either dose of TOFANIB and  $\geq 1\%$  greater than reported in patients receiving placebo are shown in Table 3.

**Table 3: Common Adverse Reactions\* in -UC Patients during the Maintenance Trial (Study UC-III)**

|                                        | <b>TOFANIB 5mg<br/>Twice Daily</b> | <b>TOFANIB 10mg<br/>Twice Daily</b> | <b>Placebo</b>       |
|----------------------------------------|------------------------------------|-------------------------------------|----------------------|
| <b>Preferred Term</b>                  | <b>N=198<br/>(%)</b>               | <b>N=196<br/>(%)</b>                | <b>N=198<br/>(%)</b> |
| Nasopharyngitis                        | 10                                 | 14                                  | 6                    |
| Elevated cholesterol levels**          | 5                                  | 9                                   | 1                    |
| Headache                               | 9                                  | 3                                   | 6                    |
| Upper respiratory tract infection      | 7                                  | 6                                   | 4                    |
| Increased blood creatine phosphokinase | 3                                  | 7                                   | 2                    |
| Rash                                   | 3                                  | 6                                   | 4                    |
| Diarrhea                               | 2                                  | 5                                   | 3                    |
| Herpes zoster                          | 1                                  | 5                                   | 1                    |
| Gastroenteritis                        | 3                                  | 4                                   | 3                    |
| Anemia                                 | 4                                  | 2                                   | 2                    |
| Nausea                                 | 1                                  | 4                                   | 3                    |

\* reported in  $\geq 4\%$  of patients treated with either dose of TOFANIB and  $\geq 1\%$  greater than reported for placebo.

\*\* includes hypercholesterolemia, hyperlipidemia, blood cholesterol increased, dyslipidemia, blood triglycerides increased, low density lipoprotein increased, low density lipoprotein abnormal, or lipids increased.

Dose-dependent adverse reactions seen in patients treated with TOFANIB 10 mg twice daily, in comparison to 5 mg twice daily, include the following: herpes zoster infections, serious infections, and NMSC.

During the UC controlled clinical studies (8-week induction and 52-week maintenance studies), which included 1220 patients, 0 cases of solid cancer or lymphoma were observed in TOFANIB -treated patients.

In the long-term extension study, malignancies (including solid cancers, lymphomas and NMSC) were observed in patients treated with TOFANIB 5 mg and 10 mg twice daily [see Warnings and Precautions (5.3)]. Five cases of pulmonary embolism were reported in patients taking TOFANIB 10 mg twice daily, including one fatality in a patient with advanced cancer.

### **Tuberculosis**

In the seven controlled trials, during the 0 to 3 months exposure, tuberculosis was not reported in patients who received placebo, 5 mg twice daily of TOFANIB, or 10 mg twice daily of TOFANIB.

In the seven controlled trials, during the 0 to 12 months exposure, tuberculosis was reported in 0 patients who received 5 mg twice daily of TOFANIB and 6 patients (0.5 events per 100 patient-years) who received 10 mg twice daily of TOFANIB. The rate difference between TOFANIB doses (and the corresponding 95% confidence interval) was 0.5 (0.1, 0.9) events per 100 patient-years for 10 mg twice daily TOFANIB minus 5 mg twice daily TOFANIB.

Cases of disseminated tuberculosis were also reported. The median TOFANIB exposure prior to diagnosis of tuberculosis was 10 months (range from 152 to 960 days).

### **Opportunistic Infections (excluding tuberculosis)**

In the seven controlled trials, during the 0 to 3 months exposure, opportunistic infections were not reported in patients who received placebo, 5 mg twice daily of TOFANIB, or 10 mg twice daily of TOFANIB.

In the seven controlled trials, during the 0 to 12 months exposure, opportunistic infections were reported in 4 patients (0.3 events per 100 patient-years) who received 5 mg twice daily of TOFANIB and 4 patients (0.3 events per 100 patient-years) who received 10 mg twice daily of TOFANIB. The rate difference between TOFANIB doses (and the corresponding 95% confidence interval) was 0 (-0.5, 0.5) events per 100 patient-years for 10 mg twice daily TOFANIB minus 5 mg twice daily TOFANIB.

The median TOFANIB exposure prior to diagnosis of an opportunistic infection was 8 months (range from 41 to 698 days).

### **Malignancy**

In the seven controlled trials, during the 0 to 3 months exposure, malignancies excluding NMSC were reported in 0 patients who received placebo and 2 patients (0.3 events per 100 patient-years) who received either TOFANIB 5 mg or 10 mg twice daily. The rate difference between treatment groups (and the corresponding 95% confidence interval) was 0.3 (-0.1, 0.7) events per 100 patient-years for the combined 5 mg and 10 mg twice daily TOFANIB group minus placebo.

In the seven controlled trials, during the 0 to 12 months exposure, malignancies excluding NMSC were reported in 5 patients (0.4 events per 100 patient-years) who received 5 mg twice daily of TOFANIB and 7 patients (0.6 events per 100 patient-years) who received 10 mg twice daily of TOFANIB. The rate difference between TOFANIB doses (and the corresponding 95% confidence interval) was 0.2 (- 0.4, 0.7) events per 100 patient-years for 10 mg twice daily TOFANIB minus 5 mg twice daily TOFANIB. One of these malignancies was a case of lymphoma that occurred during the 0 to 12 month period in a patient treated with TOFANIB 10 mg twice daily.

The most common types of malignancy, including malignancies observed during the long-term extension, were lung and breast cancer, followed by gastric, colorectal, renal cell, prostate cancer, lymphoma, and malignant melanoma.

### **Laboratory Abnormalities**

#### Lymphopenia

In the controlled clinical trials, confirmed decreases in absolute lymphocyte counts below 500 cells/mm<sup>3</sup> occurred in 0.04% of patients for the 5 mg twice daily and 10 mg twice daily TOFANIB groups combined during the first 3 months of exposure.

Confirmed lymphocyte counts less than 500 cells/mm<sup>3</sup> were associated with an increased incidence of treated and serious infections.

#### Neutropenia

In the controlled clinical trials, confirmed decreases in ANC below 1000 cells/mm<sup>3</sup> occurred in 0.07% of patients for the 5 mg twice daily and 10 mg twice daily TOFANIB groups combined during the first 3 months of exposure.

There were no confirmed decreases in ANC below 500 cells/mm<sup>3</sup> observed in any treatment group.

There was no clear relationship between neutropenia and the occurrence of serious infections.



In the long-term safety population, the pattern and incidence of confirmed decreases in ANC remained consistent with what was seen in the controlled clinical trials.

### **Liver Enzyme Elevations**

Confirmed increases in liver enzymes greater than 3 times the upper limit of normal (3x ULN) were observed in patients treated with TOFANIB. In patients experiencing liver enzyme elevation, modification of treatment regimen, such as reduction in the dose of concomitant DMARD, interruption of TOFANIB, or reduction in TOFANIB dose, resulted in decrease or normalization of liver enzymes.

In the controlled monotherapy trials (0-3 months), no differences in the incidence of ALT or AST elevations were observed between the placebo, and TOFANIB 5 mg and 10 mg twice daily groups.

In the controlled background DMARD trials (0-3 months), ALT elevations greater than 3x ULN were observed in 1.0%, 1.3% and 1.2% of patients receiving placebo, 5 mg, and 10 mg twice daily, respectively. In these trials, AST elevations greater than 3x ULN were observed in 0.6%, 0.5% and 0.4% of patients receiving placebo, 5 mg, and 10 mg twice daily, respectively.

One case of drug-induced liver injury was reported in a patient treated with TOFANIB 10 mg twice daily for approximately 2.5 months. The patient developed symptomatic elevations of AST and ALT greater than 3x ULN and bilirubin elevations greater than 2x ULN, which required hospitalizations and a liver biopsy.

In the clinical trials in psoriatic arthritis, ankylosing spondylitis, ulcerative colitis, and polyarticular course juvenile idiopathic arthritis, changes in liver enzyme tests observed with TOFANIB treatment were similar to the changes observed in clinical trials in rheumatoid arthritis where patients received background DMARDs.

### **Lipid Elevations**

In the controlled clinical trials, dose-related elevations in lipid parameters (total cholesterol, LDL cholesterol, HDL cholesterol, triglycerides) were observed at one month of exposure and remained stable thereafter. Changes in lipid parameters during the first 3 months of exposure in the controlled clinical trials are summarized below.

- Mean LDL cholesterol increased by 15% in the TOFANIB 5 mg twice daily arm and 19% in the TOFANIB 10 mg twice daily arm.
- Mean HDL cholesterol increased by 10% in the TOFANIB 5 mg twice daily arm and 12% in the TOFANIB 10 mg twice daily arm.
- Mean LDL/HDL ratios were essentially unchanged in TOFANIB -treated patients.

In a controlled clinical trial, elevations in LDL cholesterol and ApoB decreased to pretreatment levels in response to statin therapy.

In the long-term safety population, elevations in lipid parameters remained consistent with what was seen in the controlled clinical trials.

### **Serum Creatinine Elevations**

In the controlled clinical trials, dose-related elevations in serum creatinine were observed with TOFANIB treatment. The mean increase in serum creatinine was <0.1 mg/dL in the 12-month pooled safety analysis; however with increasing duration of exposure in the long-term extensions, up to 2% of patients were discontinued from TOFANIB treatment due to the protocol-specified discontinuation criterion of an

increase in creatinine by more than 50% of baseline. The clinical significance of the observed serum creatinine elevations is unknown.

#### Other Adverse Reactions

Adverse reactions occurring in 2% or more of patients on 5 mg twice daily or 10 mg twice daily TOFANIB and at least 1% greater than that observed in patients on placebo with or without DMARD are summarized in Table 4.

**Table 4: Common Adverse Reactions\* in Clinical Trials of TOFANIB for the Treatment of Rheumatoid Arthritis with or without Concomitant DMARDs (0-3 Months)**

|                                   | <b>TOFANIB 5mg<br/>Twice Daily</b> | <b>TOFANIB 10mg<br/>Twice Daily</b> | <b>Placebo</b>   |
|-----------------------------------|------------------------------------|-------------------------------------|------------------|
| Preferred Term                    | <b>N=1336 (%)</b>                  | <b>N=1349 (%)</b>                   | <b>N=809 (%)</b> |
| Upper respiratory tract infection | 4                                  | 4                                   | 3                |
| Nasopharyngitis                   | 4                                  | 3                                   | 3                |
| Diarrhea                          | 4                                  | 3                                   | 2                |
| Headache                          | 4                                  | 3                                   | 2                |
| Hypertension                      | 2                                  | 2                                   | 1                |

N reflects randomized and treated patients from the seven clinical trials.

\* Reported in  $\geq 2\%$  of patients treated with either dose of TOFANIB and  $\geq 1\%$  greater than that reported for placebo.

\*\* The recommended dose of TOFANIB for the treatment of rheumatoid arthritis is 5 mg twice daily.

Other adverse reactions occurring in controlled and open-label extension studies included:

**Blood and lymphatic system disorders:** Anemia

**Infections and infestations:** Diverticulitis

**Metabolism and nutrition disorders:** Dehydration

**Psychiatric disorders:** Insomnia

**Nervous system disorders:** Paresthesia

**Respiratory, thoracic and mediastinal disorders:** Dyspnea, cough, sinus congestion, interstitial lung disease (cases were limited to patients with rheumatoid arthritis and some were fatal)

**Gastrointestinal disorders:** Abdominal pain, dyspepsia, vomiting, gastritis, nausea

**Hepatobiliary disorders:** Hepatic steatosis

**Skin and subcutaneous tissue disorders:** Rash, erythema, pruritus

**Musculoskeletal, connective tissue and bone disorders:** Musculoskeletal pain, arthralgia, tendonitis, joint swelling

**Neoplasms benign, malignant and unspecified (including cysts and polyps):** Non-melanoma skin cancers

**General disorders and administration site conditions:** Pyrexia, fatigue, peripheral edema

#### Psoriatic Arthritis

TOFANIB 5 mg twice daily and 10 mg twice daily were studied in 2 double blind Phase 3 clinical trials in patients with active psoriatic arthritis (PsA). Although other doses of TOFANIB have been studied, the recommended dose of TOFANIB is 5 mg twice daily. A dosage of TOFANIB 10 mg twice daily is not recommended for the treatment of PsA.

Study PsA-I (NCT01877668) had a duration of 12 months and enrolled patients who had an inadequate response to a nonbiologic DMARD and who were naïve to treatment with a TNF blocker. Study PsA-I included a 3-month placebo-controlled period and also included adalimumab 40 mg subcutaneously once every 2 weeks for 12 months.

Study PsA-II (NCT01882439) had a duration of 6 months and enrolled patients who had an inadequate response to at least one approved TNF blocker. This clinical trial included a 3-month placebo controlled period.

In these combined Phase 3 clinical trials, 238 patients were randomized and treated with TOFANIB 5 mg twice daily and 236 patients were randomized and treated with TOFANIB 10 mg twice daily. All patients in the clinical trials were required to receive treatment with a stable dose of a nonbiologic DMARD [the majority (79%) received methotrexate]. The study population randomized and treated with TOFANIB (474 patients) included 45 (9.5%) patients aged 65 years or older and 66 (13.9%) patients with diabetes at baseline.

The safety profile observed in patients with active psoriatic arthritis treated with TOFANIB was consistent with the safety profile observed in rheumatoid arthritis patients.

#### Ulcerative Colitis

The following safety data were based on 4 randomized, double-blind, placebo-controlled studies: 2 Phase 3 induction studies of identical design (UC I and UC II), a Phase 3 maintenance study (UC III), and 1 dose-ranging Phase 2 induction study (UC V). Patients with moderately to severely active ulcerative colitis were enrolled in the Phase 2 and Phase 3 induction studies. In the induction studies, randomized patients received treatment with TOFANIB 10 mg twice daily (938 patients combined) or placebo (282 patients combined) for up to 8 weeks. Patients who completed either Study UC I or Study UC II and achieved clinical response entered Study UC III. In Study UC III, patients were re-randomized, such that 198 patients received TOFANIB 5 mg twice daily, 196 patients received TOFANIB 10 mg twice daily, and 198 patients received placebo for up to 52 weeks. Concomitant use of immunosuppressants or biologics was prohibited during these studies. Concomitant stable doses of oral corticosteroids were allowed in the induction studies, with taper of corticosteroids to discontinuation mandated within 15 weeks of entering the maintenance study. In addition to the induction and maintenance studies, long-term safety was evaluated in an open label long term extension study (Study UC IV).

In induction and maintenance studies, across all treatment groups, the most common categories of serious adverse reactions were gastrointestinal disorders and infections, and the most common serious adverse reaction was worsening of ulcerative colitis.

In the controlled clinical studies for ulcerative colitis, 1 case of breast cancer was reported in a placebo-treated patient and no cases of solid cancers or lymphoma were observed in TOFANIB treated patients. Malignancies have also been observed in the long term extension study in patients with ulcerative colitis treated with TOFANIB, including solid cancers and lymphoma.

In induction and maintenance studies, the most frequent reason for study discontinuation was worsening of ulcerative colitis. Excluding discontinuations due to worsening of ulcerative colitis, the proportion of patients who discontinued due to adverse reactions was less than 5% in any of the TOFANIB or placebo treatment groups in these studies.

Overall, the safety profile observed in patients with ulcerative colitis treated with TOFANIB was consistent with the safety profile of TOFANIB across indications.

## Post-marketing Experience

The following adverse reactions have been identified during post-approval use of TOFANIB. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

*Immune system disorders:* Drug hypersensitivity (events such as angioedema and urticaria have been observed).

## 4.8 Overdosage

There is no specific antidote for overdose with tofacitinib. In case of an overdose, it is recommended that the patient be monitored for signs and symptoms of adverse reactions.

In a study in subjects with end stage renal disease (ESRD) undergoing hemodialysis, plasma tofacitinib concentrations declined more rapidly during the period of hemodialysis and dialyzer efficiency, calculated as dialyzer clearance/blood flow entering the dialyzer, was high [mean (SD) = 0.73 (0.15)]. However, due to the significant non-renal clearance of tofacitinib, the fraction of total elimination occurring by hemodialysis was small, and thus limits the value of hemodialysis for treatment of overdose with TOFANIB.

## 5. DRUG INTERACTIONS

Table 5 includes drugs with clinically important drug interactions when administered concomitantly with TOFANIB and instructions for preventing or managing them.

**Table 5: Clinical Relevant Interactions Affecting TOFANIB When Co-administered with Other Drugs**

| <b>Strong CYP3A4 Inhibitors (e.g., ketoconazole)</b>                                                 |                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Clinical Impact</i>                                                                               | Increased exposure to tofacitinib                                                                                                                               |
| <i>Intervention</i>                                                                                  | Dosage adjustment of TOFANIB is recommended                                                                                                                     |
| <b>Moderate CYP3A4 Inhibitors Co-administered with Strong CYP2C19 Inhibitors (e.g., fluconazole)</b> |                                                                                                                                                                 |
| <i>Clinical Impact</i>                                                                               | Increased exposure to tofacitinib                                                                                                                               |
| <i>Intervention</i>                                                                                  | Dosage adjustment of TOFANIB is recommended                                                                                                                     |
| <b>Strong CYP3A4 Inducers (e.g., rifampin)</b>                                                       |                                                                                                                                                                 |
| <i>Clinical Impact</i>                                                                               | Decreased exposure to tofacitinib and may result in loss of or reduced clinical response                                                                        |
| <i>Intervention</i>                                                                                  | Co-administration with TOFANIB is not recommended                                                                                                               |
| <b>Immunosuppressive Drugs (e.g., azathioprine, tacrolimus, cyclosporine)</b>                        |                                                                                                                                                                 |
| <i>Clinical Impact</i>                                                                               | Risk of added immunosuppression; co-administration with biologic DMARDs or potent immunosuppressants has not been studied in patients with rheumatoid arthritis |
| <i>Intervention</i>                                                                                  | Co-administration with TOFANIB is not recommended                                                                                                               |

### *Interactions Affecting the Use of TOFANIB*

Since tofacitinib is metabolized by CYP3A4, interaction with drugs that inhibit or induce CYP3A4 is likely. Tofacitinib exposure is increased when coadministered with potent inhibitors of cytochrome P450 (CYP) 3A4 (e.g., ketoconazole) or when administration of one or more concomitant medications results in both moderate inhibition of CYP3A4 and potent inhibition of CYP2C19 (e.g., fluconazole).

Tofacitinib exposure is decreased when coadministered with potent CYP inducers (e.g., rifampin). Inhibitors of CYP2C19 alone or P-glycoprotein are unlikely to significantly alter the PK of tofacitinib.

Concomitant administration with methotrexate (15-25 mg MTX once weekly) had no effect on the PK of tofacitinib. Coadministration of ketoconazole, a strong CYP3A4 inhibitor, with a single dose of tofacitinib increased the AUC and C<sub>max</sub> by 103% and 16%, respectively. Coadministration of fluconazole, a moderate inhibitor of CYP3A4 and a strong inhibitor of CYP2C19, increased the AUC and C<sub>max</sub> of tofacitinib by 79% and 27%, respectively. Coadministration of tacrolimus (Tac), a mild inhibitor of CYP3A4, increased the AUC of tofacitinib by 21% and decreased the C<sub>max</sub> of tofacitinib by 9%. Coadministration of cyclosporine (CsA), a moderate inhibitor of CYP3A4, increased the AUC of tofacitinib by 73% and decreased C<sub>max</sub> of tofacitinib by 17%. The combined use of multiple-dose tofacitinib with these potent immunosuppressives has not been studied in patients with rheumatoid arthritis, psoriasis, psoriatic arthritis, ankylosing spondylitis, ulcerative colitis, or polyarticular course juvenile idiopathic arthritis. Coadministration of rifampin, a strong CYP3A4 inducer, decreased the AUC and C<sub>max</sub> of tofacitinib by 84% and 74%, respectively.

#### *Potential for TOFANIB to Influence the PK of Other Drugs*

In vitro studies indicate that tofacitinib does not significantly inhibit or induce the activity of the major human drug metabolizing CYPs (CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, and CYP3A4) at concentrations exceeding 80 times the steady state total C<sub>max</sub> at 5 mg and 10 mg twice daily doses in rheumatoid arthritis, psoriatic arthritis, psoriasis, ulcerative colitis, and polyarticular course juvenile idiopathic arthritis patients. These in vitro results were confirmed by a human drug interaction study showing no changes in the PK of midazolam, a highly sensitive CYP3A4 substrate, when coadministered with tofacitinib.

In vitro studies indicate that tofacitinib does not significantly inhibit the activity of the major human drug-metabolizing uridine 5'-diphospho-glucuronosyltransferases (UGTs), [UGT1A1, UGT1A4, UGT1A6, UGT1A9, and UGT2B7] at concentrations exceeding 250 times the steady state total C<sub>max</sub> at 5 mg and 10 mg twice daily doses in rheumatoid arthritis, psoriatic arthritis, psoriasis patients, ulcerative colitis patients and polyarticular course juvenile idiopathic arthritis.

In rheumatoid arthritis patients, psoriasis patients, ulcerative colitis, and polyarticular course juvenile idiopathic arthritis patients the oral clearance of tofacitinib does not vary with time, indicating that tofacitinib does not normalize CYP enzyme activity in these patients. Therefore, coadministration with tofacitinib is not expected to result in clinically relevant increases in the metabolism of CYP substrates in RA patients, psoriasis patients, ulcerative colitis, and polyarticular course juvenile idiopathic arthritis patients.

## **6. CLINICAL PHARMACOLOGY**

### **6.1 Mechanism of Action**

Tofacitinib is a Janus kinase (JAK) inhibitor. JAKs are intracellular enzymes which transmit signals arising from cytokine or growth factor-receptor interactions on the cellular membrane to influence cellular processes of hematopoiesis and immune cell function. Within the signaling pathway, JAKs phosphorylate and activate Signal Transducers and Activators of Transcription (STATs) which modulate intracellular activity including gene expression. Tofacitinib modulates the signaling pathway at the point of JAKs, preventing the phosphorylation and activation of STATs. JAK enzymes transmit cytokine signaling through pairing of JAKs (e.g., JAK1/JAK3, JAK1/JAK2, JAK1/TyK2, JAK2/JAK2). Tofacitinib inhibited the in vitro activities of JAK1/JAK2, JAK1/JAK3, and JAK2/JAK2 combinations with IC<sub>50</sub> of 406, 56, and 1377 nM, respectively. However, the relevance of specific JAK combinations to therapeutic effectiveness is not known.

### **6.2 Pharmacodynamics**

Treatment with tofacitinib was associated with dose-dependent reductions of circulating CD16/56+ natural killer cells, with estimated maximum reductions occurring at approximately 8-10 weeks after initiation of

therapy. These changes generally resolved within 2-6 weeks after discontinuation of treatment. Treatment with tofacitinib was associated with dose-dependent increases in B cell counts. Changes in circulating T-lymphocyte counts and T-lymphocyte subsets (CD3+, CD4+ and CD8+) were small and inconsistent. The clinical significance of these changes is unknown.

Total serum IgG, IgM, and IgA levels after 6-month dosing in patients with rheumatoid arthritis were lower than placebo; however, changes were small and not dose-dependent.

After treatment with tofacitinib in patients with rheumatoid arthritis, rapid decreases in serum C-reactive protein (CRP) were observed and maintained throughout dosing. Changes in CRP observed with tofacitinib treatment do not reverse fully within 2 weeks after discontinuation, indicating a longer duration of pharmacodynamic activity compared to the pharmacokinetic half-life.

Similar changes in T cells, B cells, and serum CRP have been observed in patients with active psoriatic arthritis although reversibility was not assessed. Total serum immunoglobulins were not assessed in patients with active psoriatic arthritis.

### 6.3 Pharmacokinetics

Following oral administration of tofacitinib, peak plasma concentrations are reached within 0.5-1 hour, elimination half-life is about 3 hours and a dose-proportional increase in systemic exposure was observed in the therapeutic dose range. Steady state concentrations are achieved in 24-48 hours with negligible accumulation after twice daily administration.

**Table 6: Pharmacokinetic Parameters of TOFANIB Following Multiple Oral Dosing**

| <b>PK Parameters<sup>a</sup> (CV%)</b> | <b>TOFANIB</b>                     |
|----------------------------------------|------------------------------------|
| <b>Dosing Regimen</b>                  | <b>5mg<br/>Twice Daily</b>         |
| AUC <sub>24</sub> (ng.h/mL)            | 263.4 (15)                         |
| C <sub>max</sub> (ng/mL)               | 42.7 (26)                          |
| C <sub>min</sub> (ng/mL)               | 1.41 (40)                          |
| T <sub>max</sub> (hours)               | 1.0<br>(0.5 to 14.0 <sup>b</sup> ) |

Abbreviations: AUC<sub>24</sub> = area under the concentration-time profile from time 0 to 24 hours; C<sub>max</sub> = maximum plasma concentration; C<sub>min</sub> = minimum plasma concentration; T<sub>max</sub> = time to C<sub>max</sub>; CV = Coefficient of variation.

a Values represent the geometric mean, except T<sub>max</sub>, for which the median (range) is shown.

b Values beyond 12 hours were after the evening dose which was administered 12 hours after the morning dose of twice-daily TOFANIB.

### Absorption

The absolute oral bioavailability of tofacitinib is 74%. Co-administration of tofacitinib with a high-fat meal resulted in no changes in AUC while C<sub>max</sub> was reduced by 32%. In clinical trials, tofacitinib was administered without regard to meals.

### Distribution

After intravenous administration, the volume of distribution is 87 L. The protein binding of tofacitinib is approximately 40%. Tofacitinib binds predominantly to albumin and does not appear to bind to  $\alpha$ 1- acid glycoprotein. Tofacitinib distributes equally between red blood cells and plasma.

### Metabolism and Excretion

Clearance mechanisms for tofacitinib are approximately 70% hepatic metabolism and 30% renal excretion of the parent drug. The metabolism of tofacitinib is primarily mediated by CYP3A4 with minor contribution from CYP2C19. In a human radiolabeled study, more than 65% of the total circulating radioactivity was accounted for by unchanged tofacitinib, with the remaining 35% attributed to 8 metabolites, each accounting for less than 8% of total radioactivity. The pharmacologic activity of tofacitinib is attributed to the parent molecule.

### Population Pharmacokinetics in Rheumatoid Arthritis Patients

Population pharmacokinetic analyses indicated that the coefficient of variation (%) in AUC of tofacitinib for patients with rheumatoid arthritis was 22.0% and psoriatic arthritis was 34%. (Table 6).

**Table 7. TOFANIB Exposure in Patient Populations at 5 mg Twice Daily Dose**

| Pharmacokinetic Parameters<br>Geometric Mean (CV%) <sup>a</sup> | TOFANIB 5mg Twice daily |
|-----------------------------------------------------------------|-------------------------|
|                                                                 | Rheumatoid Arthritis    |
| AUC <sub>0-24,ss</sub> (ng.h/mL)                                | 504<br>(22.0%)          |
| AUC <sub>0-24,ss</sub> (ng.h/mL)                                | 419<br>(34.1%)          |

Abbreviations: AUC<sub>0-24,ss</sub>=area under the plasma concentration-time curve over 24 hours at steady state; CV=coefficient of variation.

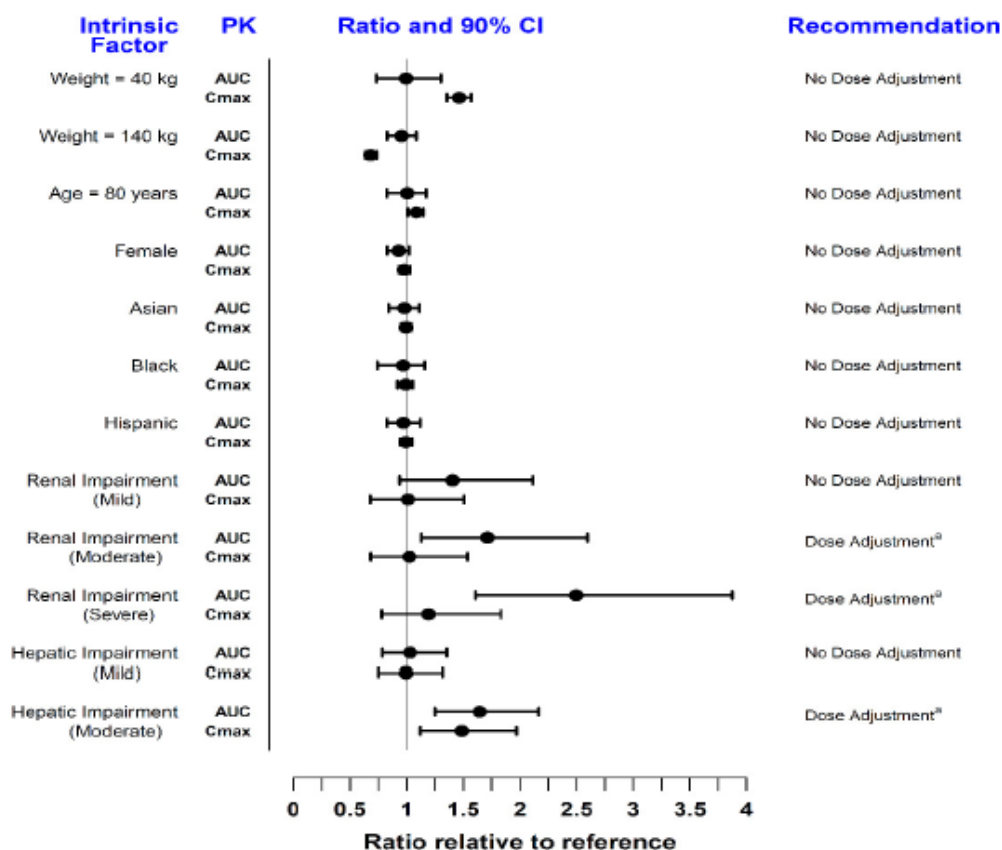
<sup>a</sup> Pharmacokinetic parameters estimated based on population pharmacokinetic analysis

### Specific Populations

Covariate evaluation as part of population PK analyses in patient populations indicated no clinically relevant change in tofacitinib exposure, after accounting for differences in renal function (i.e., creatinine clearance) between patients, based on age, weight, gender and race. An approximately linear relationship between body weight and volume of distribution was observed, resulting in higher peak (C<sub>max</sub>) and lower trough (C<sub>min</sub>) concentrations in lighter patients. However, this difference is not considered to be clinically relevant.

The effect of renal and hepatic impairment and other intrinsic factors on the pharmacokinetics of tofacitinib is shown in Figure 1.

**Figure 1: Impact of Intrinsic Factors on Tofacitinib Pharmacokinetics**



Note: Reference values for weight, age, gender, and race comparisons are 70 kg, 55 years, male, and white, respectively; reference groups for renal and hepatic impairment data are subjects with normal renal and hepatic function.

<sup>a</sup> (Refer to section 4.2 *Posology and method of administration*) for dosage adjustment in RA, PsA patients.

In subjects with ESRD maintained on hemodialysis, mean AUC was approximately 40% higher compared with historical healthy subject data, consistent with approximately 30% contribution of renal clearance to the total clearance of tofacitinib. Dose adjustment is recommended in RA, PsA patients with ESRD maintained on hemodialysis.

### Pharmacokinetics in Patients with Active Ulcerative Colitis

Population PK analysis in ulcerative colitis patients indicated that there were no clinically relevant differences in tofacitinib exposure (AUC), based on age, weight, gender, and race. Exposure in women was 15% higher than in men, and Asian patients had 7.3% higher exposure than non-Asian. Volume of distribution increased with body weight resulting in higher peak ( $C_{max}$ ) and lower trough ( $C_{min}$ ) concentrations in lighter patients. However, this difference is not considered to be clinically relevant. The between-subject variability (% coefficient of variation) in AUC of tofacitinib is estimated to be approximately 23% and 25% at the 5 mg twice daily dose and 10 mg twice daily dose, respectively, in ulcerative colitis patients.



## Drug Interaction Studies

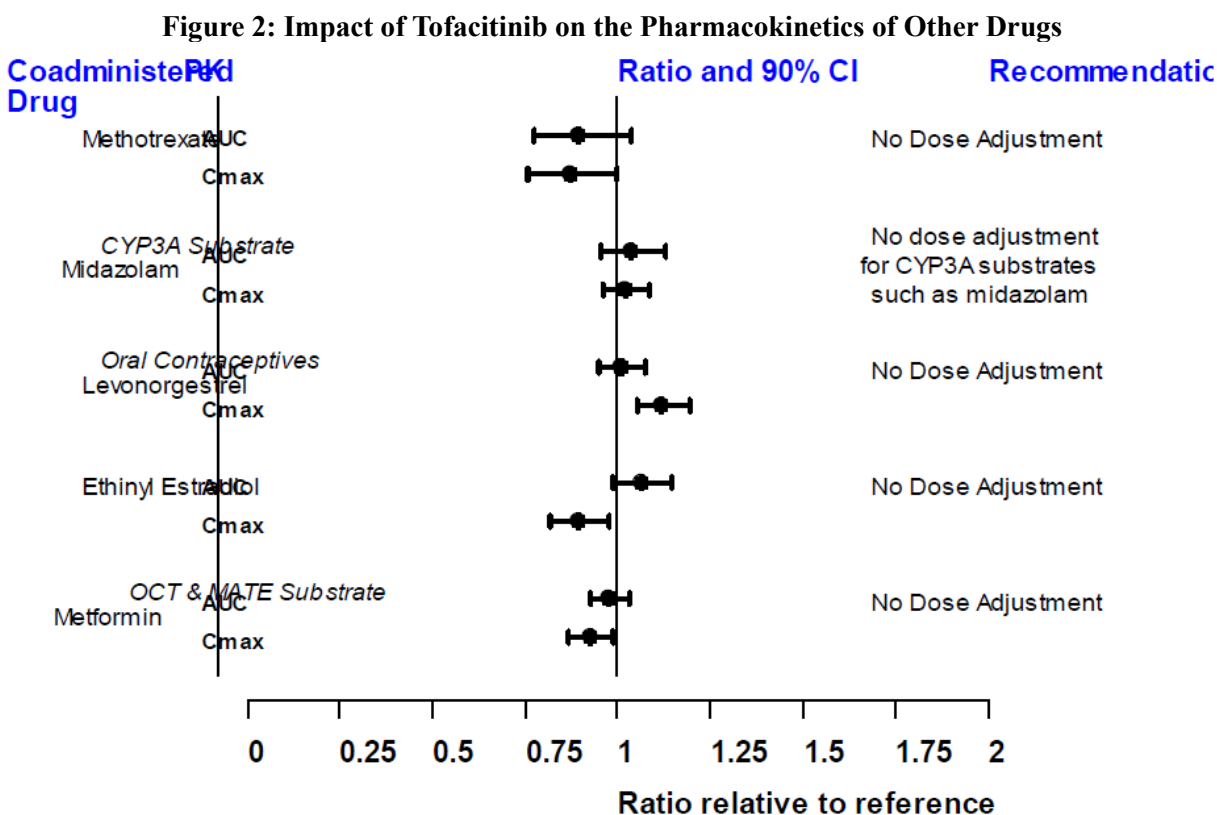
### Potential for Tofacitinib to Influence the Pharmacokinetics of Other Drugs

*In vitro* studies indicate that tofacitinib does not significantly inhibit or induce the activity of the major human drug-metabolizing CYPs (CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, and CYP3A4) at concentrations exceeding 160 times the steady state C<sub>max</sub> of a 5 mg twice daily dose. These *in vitro* results were confirmed by a human drug interaction study showing no changes in the pharmacokinetics of midazolam, a highly sensitive CYP3A4 substrate, when co-administered with tofacitinib.

In rheumatoid arthritis patients, the oral clearance of tofacitinib does not vary with time, indicating that tofacitinib does not normalize CYP enzyme activity in rheumatoid arthritis patients. Therefore, co-administration with tofacitinib is not expected to result in clinically relevant increases in the metabolism of CYP substrates in rheumatoid arthritis patients.

*In vitro* data indicate that the potential for tofacitinib to inhibit transporters such as P-glycoprotein, organic anionic or cationic transporters at therapeutic concentrations is low.

Dosing recommendations for co-administered drugs following administration with tofacitinib are shown in Figure 2.

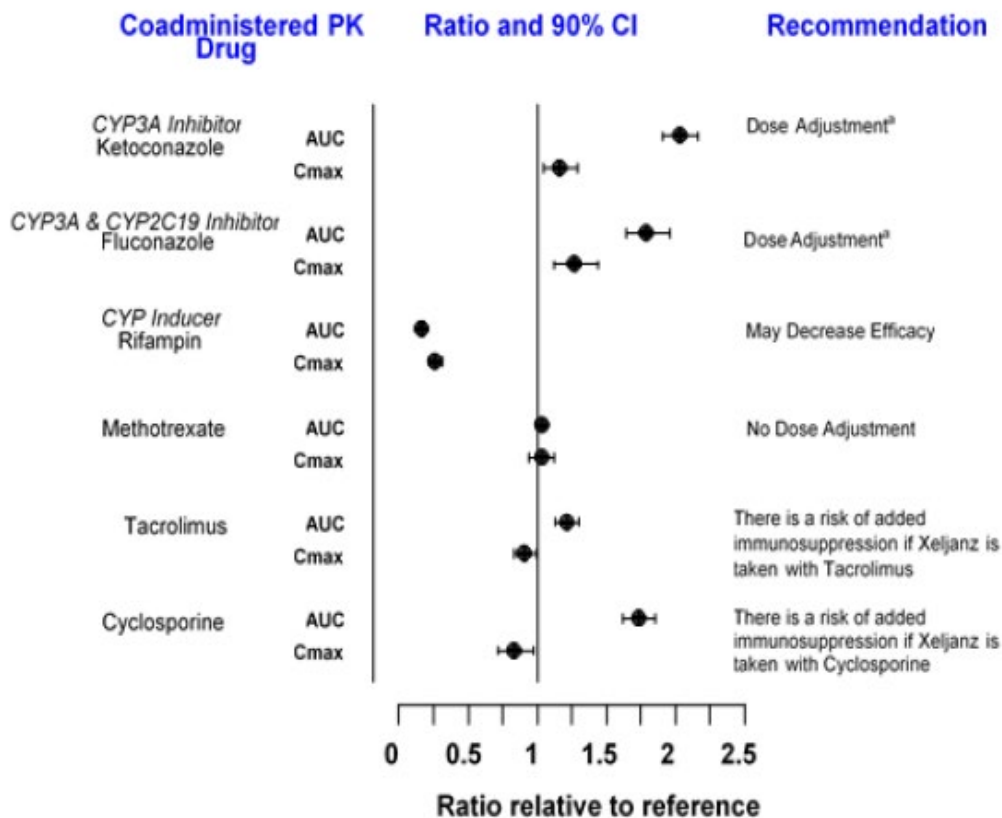


Note: Reference group is administration of concomitant medication alone; OCT = Organic Cationic Transporter; MATE = Multidrug and Toxic Compound Extrusion

### Potential for Other Drugs to Influence the Pharmacokinetics of Tofacitinib

Since tofacitinib is metabolized by CYP3A4, interaction with drugs that inhibit or induce CYP3A4 is likely. Inhibitors of CYP2C19 alone or P-glycoprotein are unlikely to substantially alter the pharmacokinetics of tofacitinib. (see Figure 3).

**Figure 3: Impact of Other Drugs on the Pharmacokinetics of Tofacitinib**



Note: Reference group is administration of tofacitinib alone

<sup>a</sup> (see sections 4.2 and 5).

## 7. PHARMACEUTICAL PARTICULARS

### 7.1 List of excipients

Microcrystalline cellulose,  
lactose monohydrate,  
croscarmellose sodium,  
magnesium stearate

*Film Coat for 5 mg tablets: Opadry® Complete Film Coating System 03F180011- CN White containing:*

Titanium dioxide  
Hypromellose  
Macrogol

### 7.2 Incompatibilities

Not applicable

**7.3 Special precautions for storage**

Store below 30°C.

**7.4 Nature and contents of container**

Each foil/foil blister contains 14 film coated tablets. 4 such blisters are packed in an outer carton.

**7.5 Manufacturer**

Yangtze River Pharmaceutical Group Jiangsu Zilong Pharmaceutical Co., Ltd.  
No. 9 Yunhe West Road, Xuejia Town, Xinbei District,  
Changzhou City, Jiangsu Province, China.

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